Lakshya JEE 2.0 2025

Maths

DPP: 2

Relations and Functions

Q1 Let $A=\{1,2,3\}, B=\{1,3,5\}$. A relation $R: A \rightarrow B$ is defined by $R=\{(1,3),(1,5),(2,1)\}$. Then R^{-1} is defined by:
(A) $\{(1,2),(3,1),(1,3),(1,5)\}$
(B) $\{(1,2),(3,1),(2,1)\}$
(C) $\{(1,2),(5,1),(3,1)\}$
(D) None of these

Q2 The relation R is defined on the set of natural numbers as $\{(a,b): a=2b\}$. Then R^{-1} is given by
(A) $\{(2,1),(4,2),(6,3) \ldots\ldots\}$
(B) $\{(1,2),(2,4),(3,6) \ldots\ldots\}$
(C) R^{-1} is not defined
(D) None of these

Q3 Find the range of the relation R given by R
$=\{$
$(x,y): x \in N, y \in N$ and $x^2+y^2 \leq 24\}$

Q4 For real number x and y, we write $xRy \Leftrightarrow x-y+\sqrt{2}$ is an irrational number. Then the relation R is.
(A) Reflexive
(B) Symmetric
(C) transitive
(D) None of these

Q5 The number of reflexive relations on a set with four elements is equal to:
(A) 2^{16}
(B) 2^{4}
(C) 2^{8}
(D) 2^{12}

Q6 Let R be the real line. Consider the following subsets of the plane $R \times R$
$\mathrm{S}=\{(x,y): y=x+1$ and $0<x<2\}$
$\mathrm{T}=\{(x,y): x-y$ is an integer $\}$
Which of the following is true?
(A) Both S and T are equivalence relations on R
(B) S is equivalence relation but T is not
(C) T is an equivalence relation but S is not
(D) Neither S nor T are equivalence relations

Q7 Let P be the relation defined on the set of all real number such that $P=\left\{(a,b): \sec^2 a-\tan^2 b=1\right\}$. Then P is
(A) reflexive and symmetric but not transitive
(B) symmetric and transitive but not reflexive
(C) reflexive and transitive but not symmetric
(D) an equivalence relation

Q8 Let
$R=\{(3,3),(6,6),(9,9),(12,12),(6,12),$
$(3,9),(3,12),(3,6)\}$
be a relation on the set $A=\{3,\ 6,\ 9,\ 12\}$. The relation is
(A) reflexive and transitive only
(B) reflexive only
(C) reflexive and symmetric only
(D) equivalence relation

Q9 Let N denotes the set of all natural numbers and R be the relation on $N \times N$ defined by $(a,b)R(c,d)$ if $ad(b+c)=bc(a+d)$, then R is:
(A) Symmetric only
(B) Reflexive only

(C) Transitive only.
(D) An equivalence relation

Q10 Let L be the set of all straight lines in a plane. l_1 and l_2 are two lines in the set. R_1, R_2 and R_3 are defined relations.
(i) $l_1 R_1 l_2 : l_1$ is parallel to l_2
(ii) $l_1 R_2 l_2 : l_1$ is perpendicular to l_2
(iii) $l_1 R_3 l_2 : l_1$ intersects l_2.
Then which of the following is true?
(A) R_1, R_2 and R_3 are equivalence
(B) R_1 is equivalence
(C) R_2 and R_3 are reflexive
(D) R_1, R_2 and R_3 are not symmetric

Q11 Suppose $\mathrm{M} = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d, \in R \right\}$ and $\mathrm{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ Define a relation $\sim$ on M as follows: $\mathrm{A}, \mathrm{B} \in \mathrm{M}, \mathrm{A} \sim \mathrm{B}$ if $\mathrm{AB} = \mathrm{I}_2$., then
(A) $\sim$ is symmetric only
(B) $\sim$ is symmetric and transitive
(C) $\sim$ is reflexive and symmetric
(D) $\sim$ is an equivalence relation

Q12 Domain of the function $\frac{1}{\sqrt{x^2-1}}$ is:
(A) $(-\infty, -1) \cup (1, \infty)$
(B) $(-\infty, -1] \cup (1, \infty)$
(C) $(-\infty, -1) \cup [1, \infty)$
(D) None of these

Q13 The domain of the function $f(x) = \frac{1}{\sqrt{|x|-x}}$ is:
(A) R^+ (B) R^-
(C) R_0 (D) R

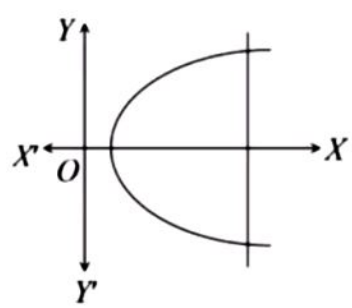
Q14 The domain of the function $f(x) = \sqrt{(2-2x-x^2)}$ is:
(A) $-3 \le x \le \sqrt{3}$
(B) $-1-\sqrt{3} \le x \le -1+\sqrt{3}$
(C) $-2 \le x \le 2$
(D) None of these

Q15 The domain of the function $f(x) = \sqrt{x-x^2} + \sqrt{4+x} + \sqrt{4-x}$ is:
(A) $[-4, \infty)$ (B) $[-4, 4]$
(C) $[0, 4]$ (D) $[0, 1]$

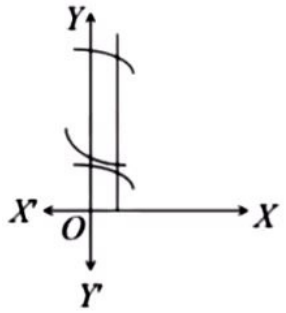
Q16 The domain of the function $\sqrt{\log(x^2-6x+6)}$ is:
(A) $(-\infty, \infty)$
(B) $(-\infty, 3-\sqrt{3}) \cup (3+\sqrt{3}, \infty)$
(C) $(-\infty, 1] \cup [5, \infty)$
(D) $[0, \infty)$

Q17 Domain of definition of the function $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$, is:
(A) $(1, 2)$
(B) $(-1, 0) \cup (1, 2)$
(C) $(1, 2) \cup (2, \infty)$
(D) $(-1, 0) \cup (1, 2) \cup (2, \infty)$

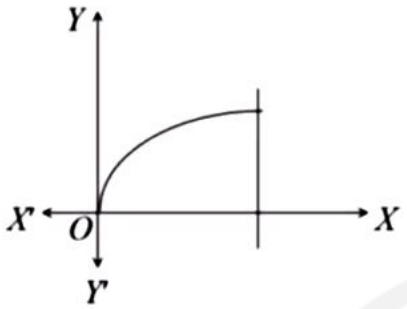
Q18 The domain of the function $f(x) = \log_{3+x}(x^2 - 1)$ is:
(A) $(-3, -1) \cup (1, \infty)$
(B) $[-3, -1) \cup [1, \infty)$
(C) $(-3, -2) \cup (-2, -1) \cup (1, \infty)$
(D) $[-3, -2) \cup (-2, -1) \cup [1, \infty)$

Q19 Which of the following can be graph of a function?
(A)


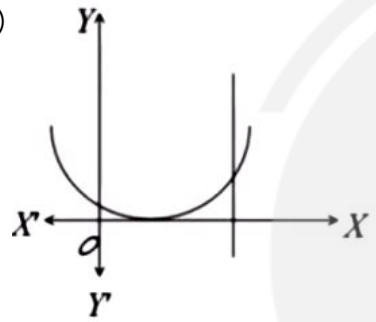
(B)



(C)



(D)



Q20 The maximum number of values of x if $|x-2|+|x-4|=2$ is:

(A) 1

(B) 2

(C) 3

(D) infinitely many

Answer Key

Q1	(C)
Q2	(B)
Q3	Range $(\mathrm{R}) = \{1, 2, 3, 4\}$
Q4	(A)
Q5	(D)
Q6	(C)
Q7	(B)
Q8	(A)
Q9	(D)
Q10	(B)
Q11	(A)
Q12	(A)
Q13	(B)
Q14	(B)
Q15	(D)
Q16	(C)
Q17	(D)
Q18	(C)
Q19	(C, D)
Q20	(D)

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